UC3M



PACKAGE REFERENCE DOCUMENT

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1 Code display packages

2 Graphics packages

2.1 TikZ – Block Diagrams

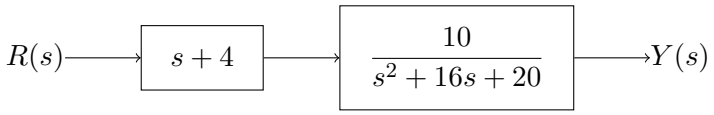
For control-system block diagrams, TikZ requires:

```
\usepackage{tikz}
\usetikzlibrary{positioning, calc}
```

2.1.1 Basic Block Diagram

```
\begin{tikzpicture}
    \node[input] (input) {$R(s)$};
    \node[block, right=of input] (G1) {$s+4$};
    \node[block, right=of G1] (G2)
        {$\dfrac{10}{s^2+16s+20}$};
    \node[output, right=of G2] (output) {$Y(s)$};

    \draw[->] (input) -- (G1);
    \draw[->] (G1) -- (G2);
    \draw[->] (G2) -- (output);
\end{tikzpicture}
```



2.1.2 Node Positioning

```
right=of node
left=of node
above=of node
below=of node

right=1cm of node
below=2\dnd cm of node % 2 times the default node distance
```

The first form uses the default `node distance`; the second overrides it locally.

2.1.3 Node Types

```
\node[input]   (input) {$R(s)$};

\node[block]   (G) {$G(s)$};

\node[sum]     (sum1) {};

\node[pickoff] (p1) {};

\node[output]  (output) {$Y(s)$};
```

2.1.4 Connections

```
% Straight line
\draw[->] (A) -- (B);

% Straight line without arrow
\draw[-] (A) -- (B);

% Vertical, then horizontal
\draw[->] (A) |- (B);

% Horizontal, then vertical
\draw[->] (A) -| (B);
```

Node anchors can be selected explicitly:

```
node.north
node.south
node.east
node.west
```

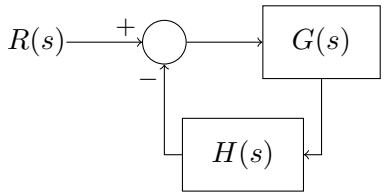
2.1.5 Summing Junctions

Like a connection, but with a plus or minus sign at the junction. The following code snippet will be used after positioning the nodes:

```
\node[sum] (sum1) {};

\draw[->] (G)
    -- node[plus, above left] {}
    (sum1);

\draw[->] (H)
    -| node[minus, below left] {}
    (sum1.south);
```



2.1.6 Pickoff Points

```
\node[pickoff, right=of G] (p1) {};

\draw[-]  (G) -- (p1);
\draw[->] (p1) -- (output);
\draw[->] (p1.south) |- (feedback.east);
```

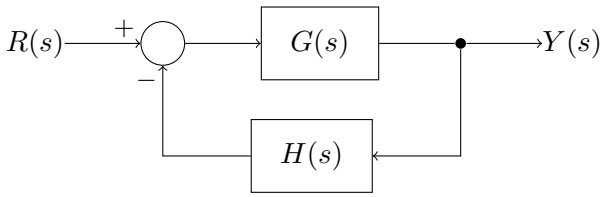
2.1.7 Feedback Loops

```
\coordinate (mid)
    at ($(sum1)!0.5!(pickoff1)$);

\node[block, below=of mid] (feedback)
    {$H(s)$};

\draw[->] (pickoff1.south)
    |- (feedback.east);

\draw[->] (feedback.west)
    -| node[minus, below left] {}
    (sum1.south);
```



2.1.8 Useful Coordinates

Point halfway between two nodes:

```
\coordinate (mid)
    at ($(A)!0.5!(B)$);
```

Other positions:

```
($(A)!0.25!(B)$)   % 25 %
($(A)!0.50!(B)$)   % 50 %
($(A)!0.75!(B)$)   % 75 %
```

Where A and B are the names of the nodes. The `calc` library must be loaded for this to work.

2.1.9 Scaling

In case the diagram doesn't fit in the page, we can scale it down using the `scale` option in the `tikzpicture` environment:

```
\begin{tikzpicture}[scale=0.8, transform shape]
    ...
\end{tikzpicture}
```

3 Page layout and formatting packages

4 Mathematical typesetting packages

5 Text formatting packages